

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 30

Title: Heme degradation and Jaundice

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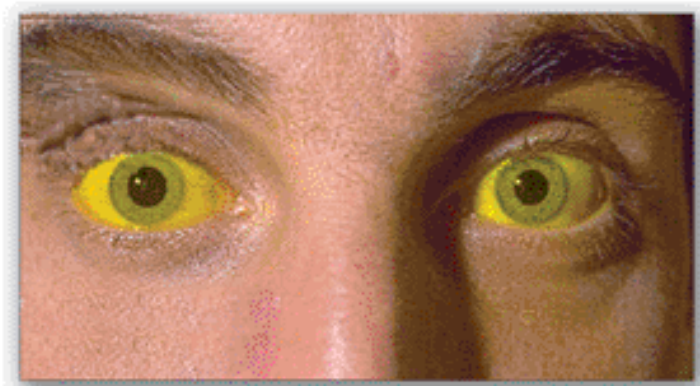
Objectives: Heme degradation and Jaundice

SOM. MK.I.BPM2.3.DM.4.BCHM.1277	Outline steps in degradation of heme to bilirubin in macrophages. Review the role of albumin in transport of bilirubin.
SOM. MK.I.BPM2.3.DM.4.BCHM.1278	Explain the consequence of a defect in uptake and conjugation of bilirubin in liver.
SOM. MK.I.BPM2.3.DM.4.BCHM.1279	Describe the fate of conjugated bilirubin in intestine. Review clinical significance of stercobilin and urobilin.
SOM. MK.I.BPM2.3.DM.4.BCHM.1280	Describe the conversion of heme to stercobilin and urobilin. Identify various tissues/organs involved in heme catabolism.
SOM. MK.I.BPM2.3.DM.4.BCHM.1281	Compare and contrast conjugated and unconjugated bilirubin with reference to chemical composition, water solubility, tissue deposition, excretion in urine and common clinical conditions where they are elevated
SOM. MK.I.BPM2.3.DM.4.BCHM.1282	Distinguish conjugated (direct) and unconjugated hyperbilirubinemia (indirect) using the Van den Bergh test.
SOM. MK.I.BPM2.3.DM.4.BCHM.1283	Explain mechanisms involved in development of physiological jaundice of newborn
SOM. MK.I.BPM2.3.DM.4.BCHM.1284	Explain rationale of phototherapy and phenobarbital in therapy for premature babies with jaundice.
SOM. MK.I.BPM2.3.DM.4.BCHM.1285	Distinguish prehepatic, hepatic and post hepatic jaundice based on laboratory findings and clinical features.
SOM. MK.I.BPM2.3.DM.4.BCHM.1286	Identify two disorders for prehepatic (hemolytic), hepatic and obstructive jaundice.
SOM. MK.I.BPM2.3.DM.4.BCHM.1287	Differentiate between inherited disorders: Gilbert syndrome, Dubin Johnson syndrome and Crigler-Najjar syndromes I and II based on age of presentation, pathogenetic mechanisms and metabolic alterations

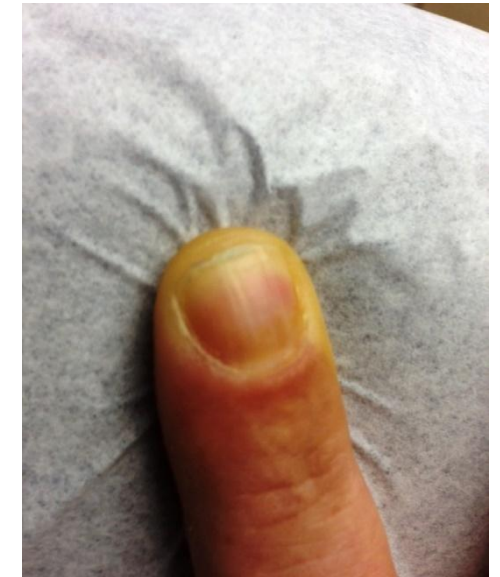
Recommended Reading

- Lippincott's Biochemistry 9th Edition
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- Lippincott's Illustrated reviews: Biochemistry [Amino Acids: Conversion to Specialized Products | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#) Section: Porphyrin metabolism; **Concept map: Fig 21.20**
- **Lippincott's Study Questions: 21.1 - 21.6**

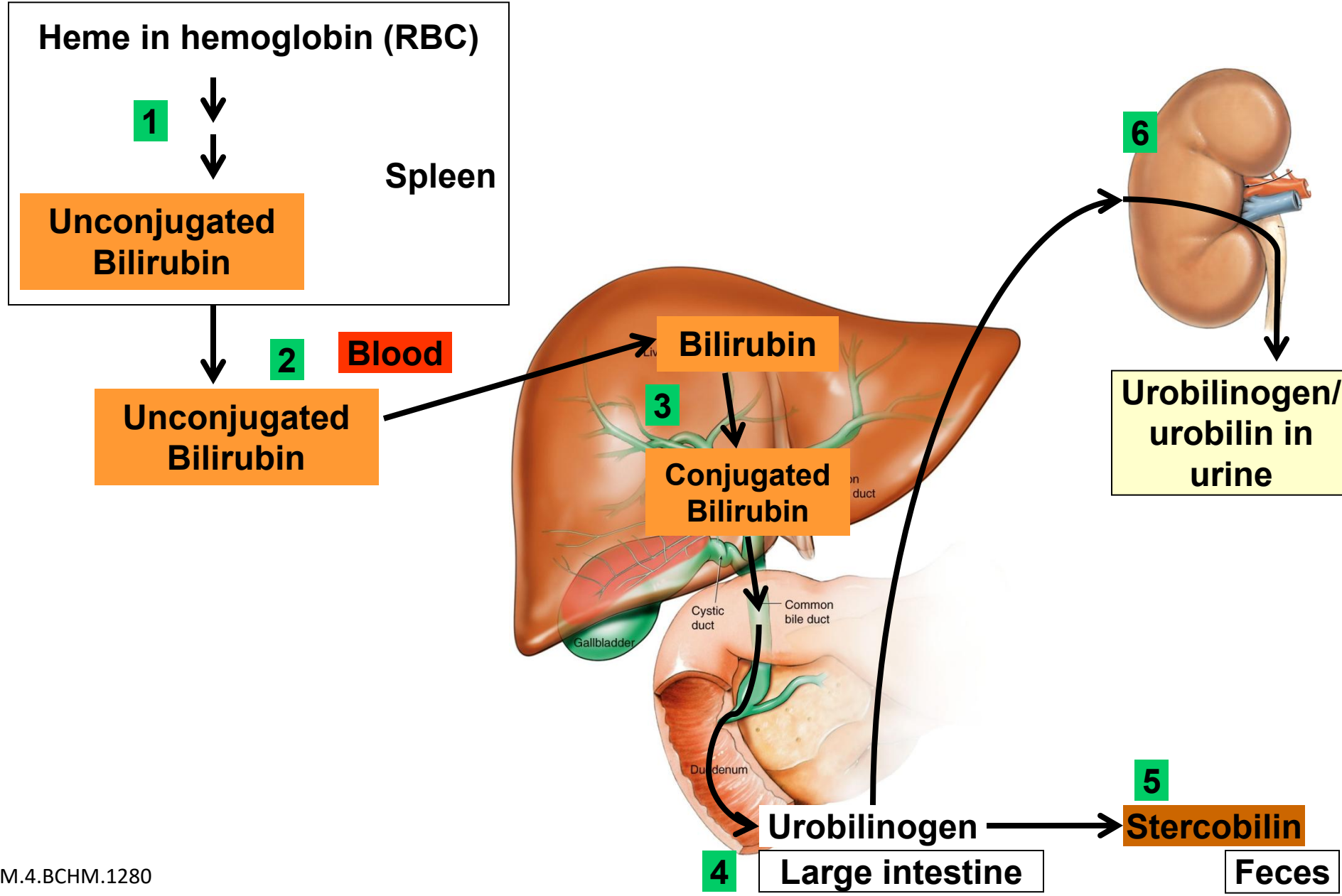
Jaundice??



- Clinical sign characterized by yellowish discoloration of skin, mucous membranes, sclera, and nail beds.
- Binding of **bilirubin (Bile pigment)** to connective tissue
- Occurs when serum bilirubin levels are greater than 2 mg/dL (hyperbilirubinemia)
- **Bile acids vs bile pigment**
 - Remember, bile salts/bile acids (required for lipid digestion and absorption) are different from bile pigments/ bilirubin (excretory waste products of bilirubin metabolism) – both are components of BILE



Bilirubin formation and excretion

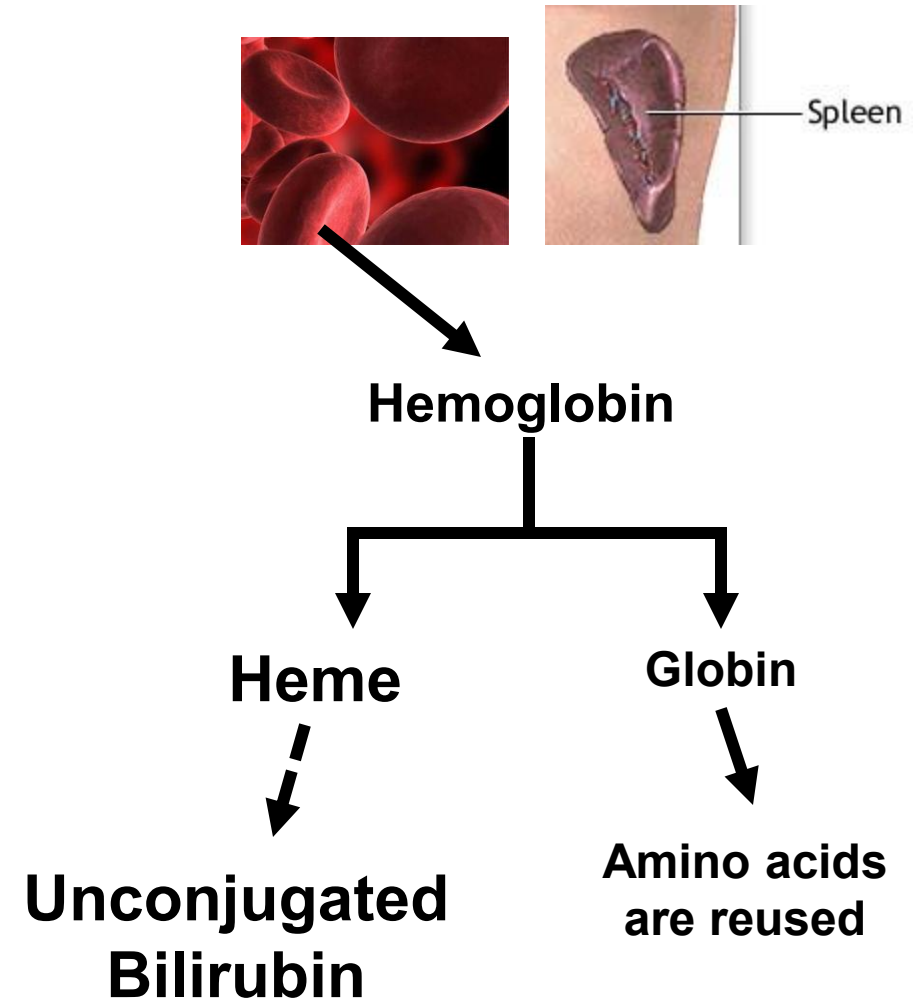


Stages of heme degradation

1. Formation of unconjugated bilirubin from heme in reticuloendothelial system
2. Transport of unconjugated bilirubin in blood from RES to liver
3. Uptake and conjugation in liver and secretion into bile
4. Formation of urobilinogen in large intestine
5. Formation of stercobilin and loss in feces
6. Absorption of urobilinogen and excretion of urobilinogen in urine

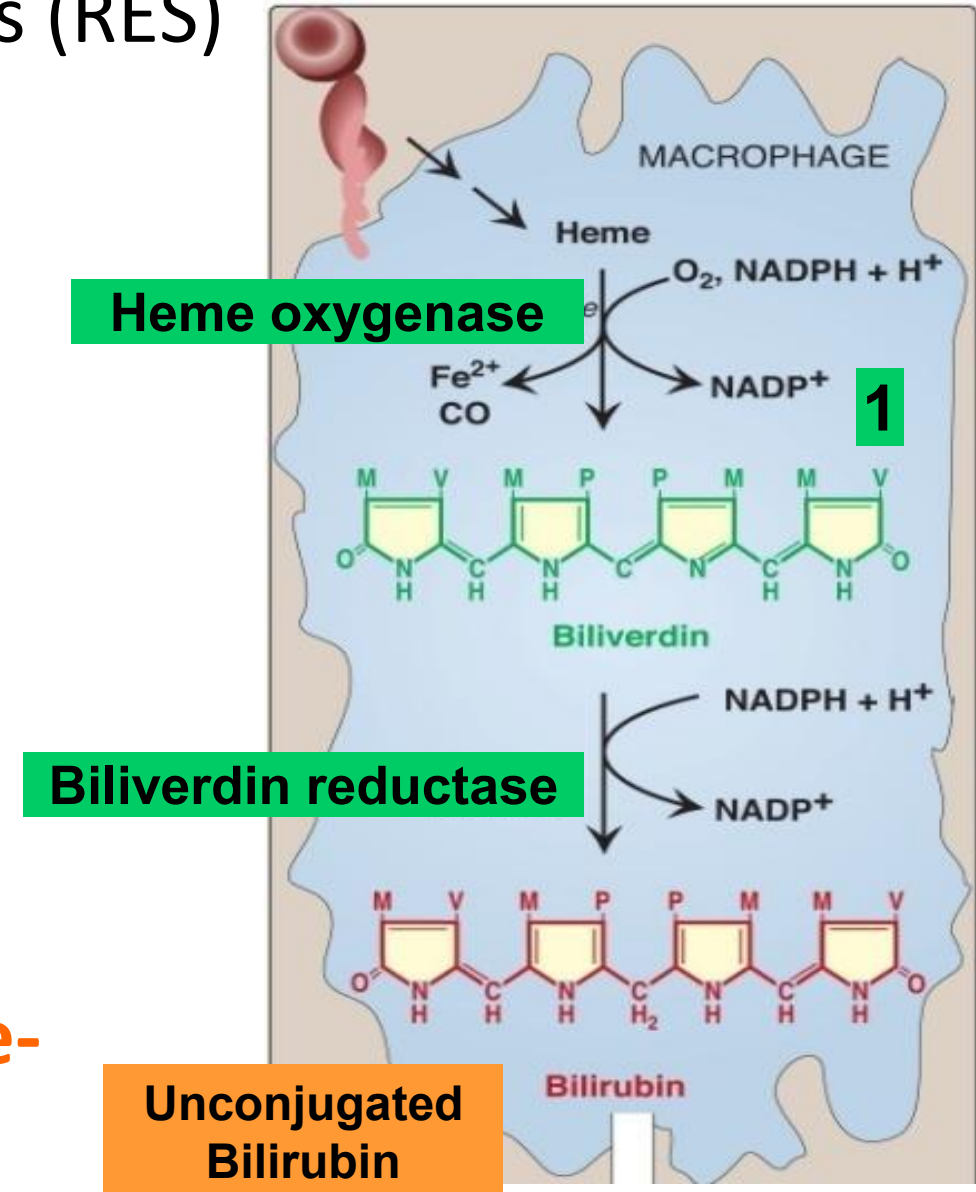
Heme degradation

- RBC lifespan about 120 days
- **Reticuloendothelial system** (Spleen, Kupffer cells of liver, macrophages) removes old RBC
- **Hemoglobin** = Heme + Globin (amino acid pool)
- Heme proteins like myoglobin and cytochromes release heme and form bilirubin



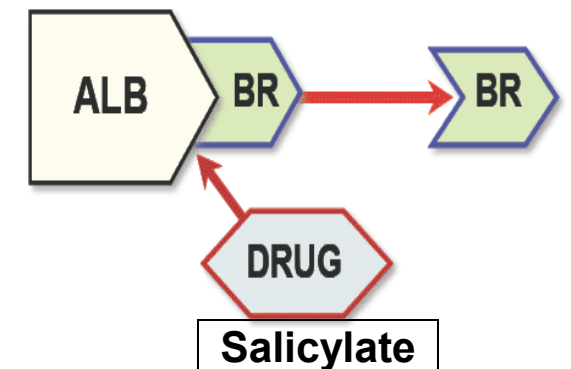
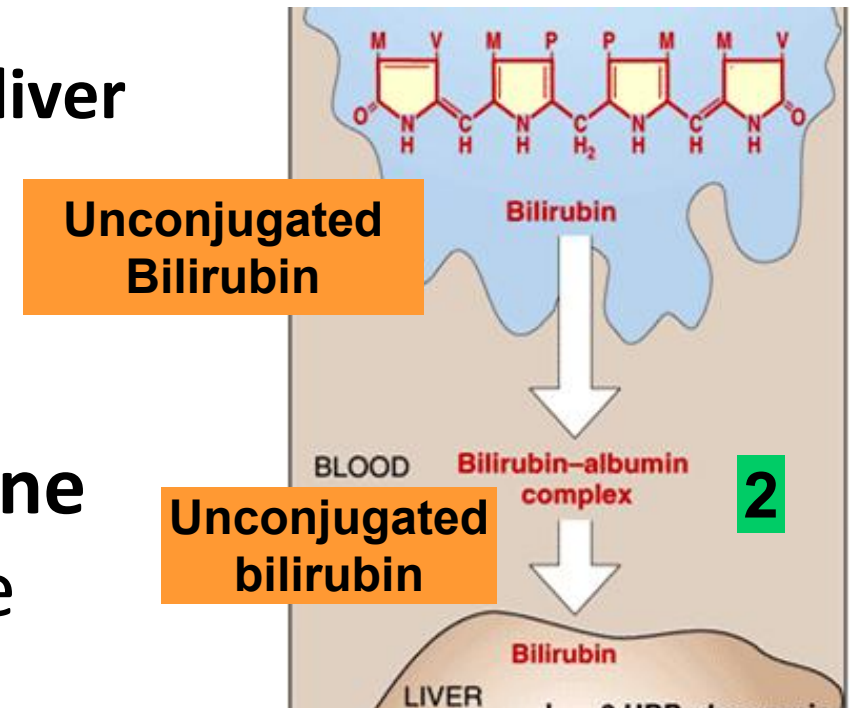
Bilirubin formation in macrophages (RES)

- Heme oxygenase
 - Fe^{2+} released
 - Porphyrin ring opens
 - CO formed
 - Biliverdin: Green pigment
- Biliverdin reductase
 - **Bilirubin (bile pigment): Orange-yellow color**



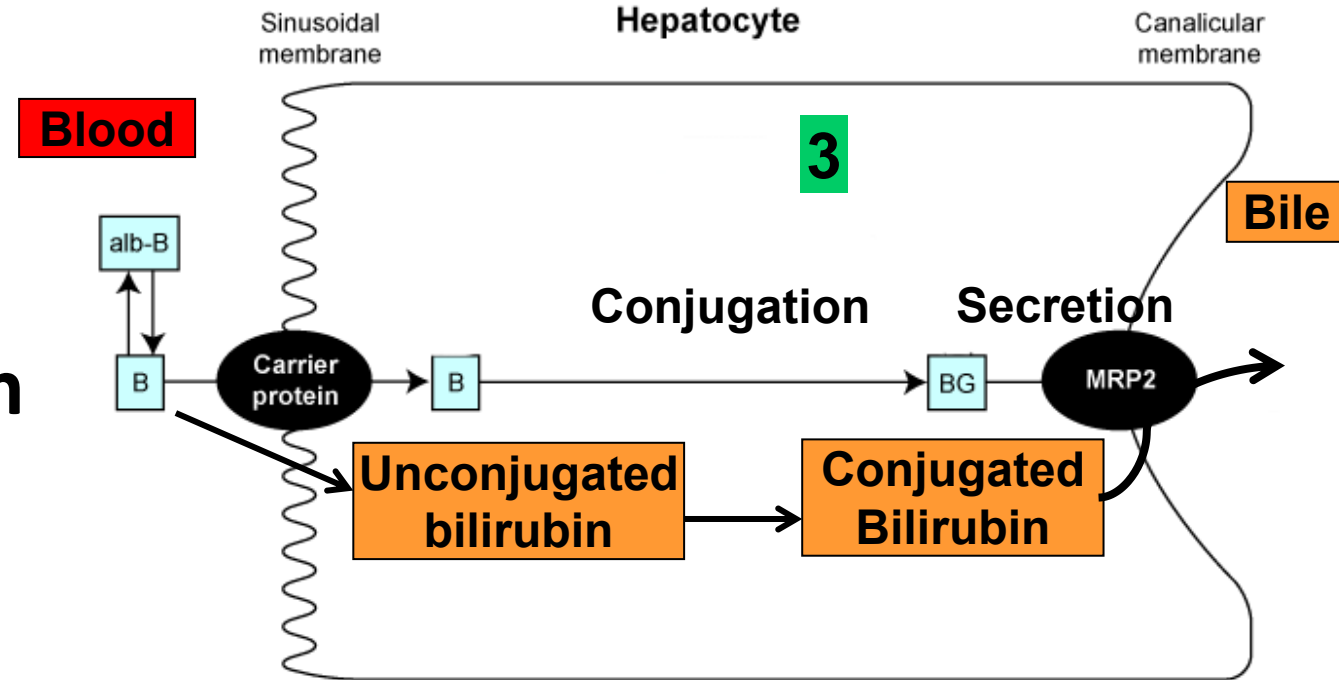
Transport of unconjugated bilirubin in blood to liver

- **Unconjugated** (NOT water soluble)
- **Albumin-bound** for transport in blood
- **Albumin binding prevents excretion in urine**
- Drugs (salicylates, sulfonamides...) displace bilirubin from albumin
 - Free unconjugated bilirubin crosses blood-brain barrier and causes **kernicterus** in infants

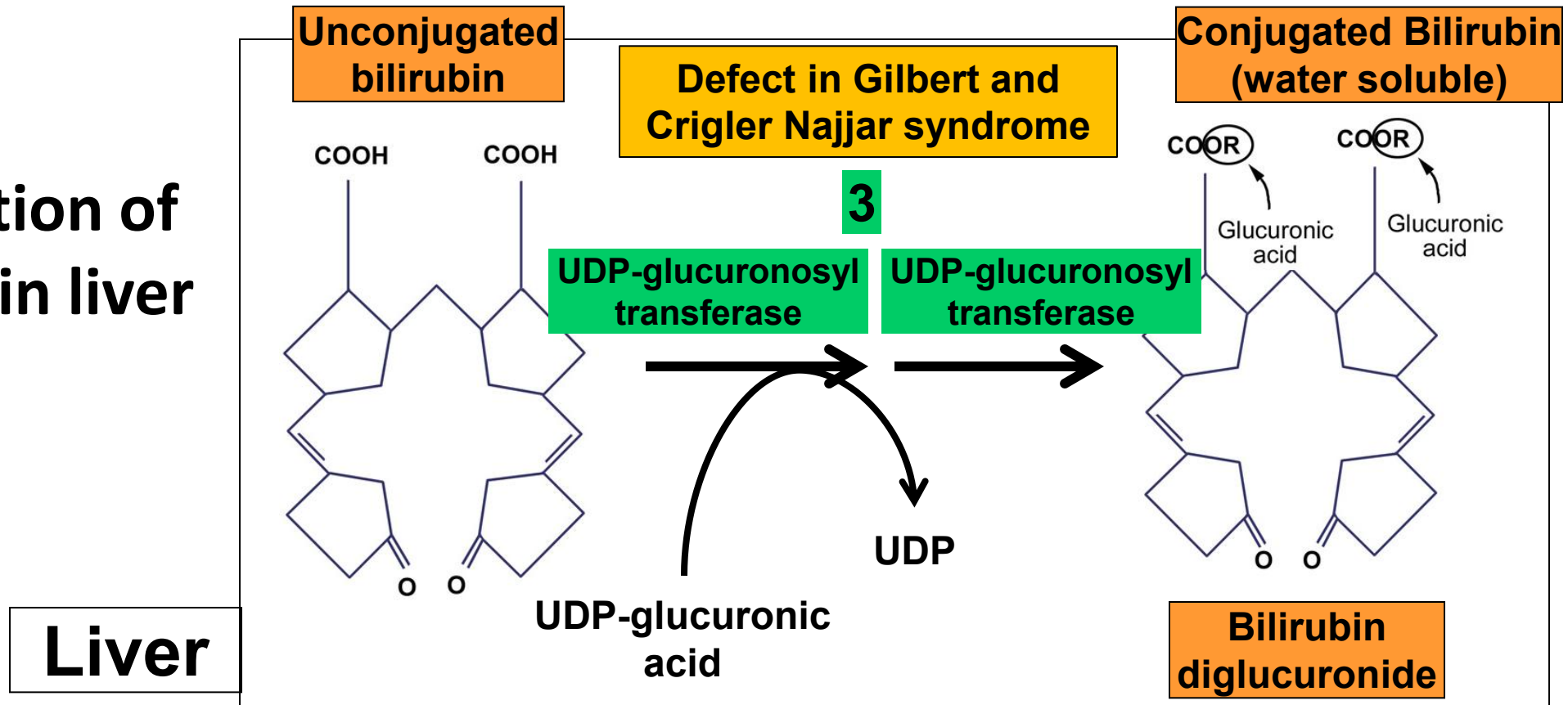


Bilirubin metabolism in liver

- Uptake of Unconjugated bilirubin by liver
- **Conjugated bilirubin formed in hepatocytes**
- **Conjugated bilirubin secreted into biliary canaliculi**



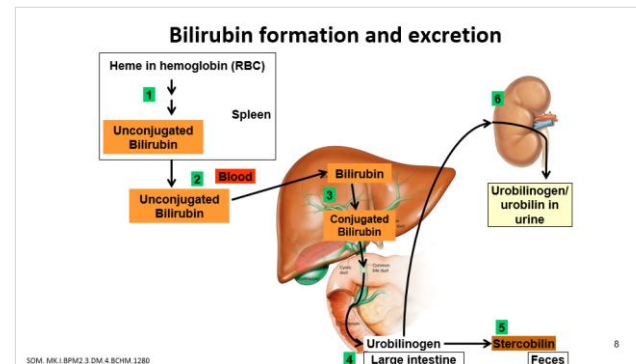
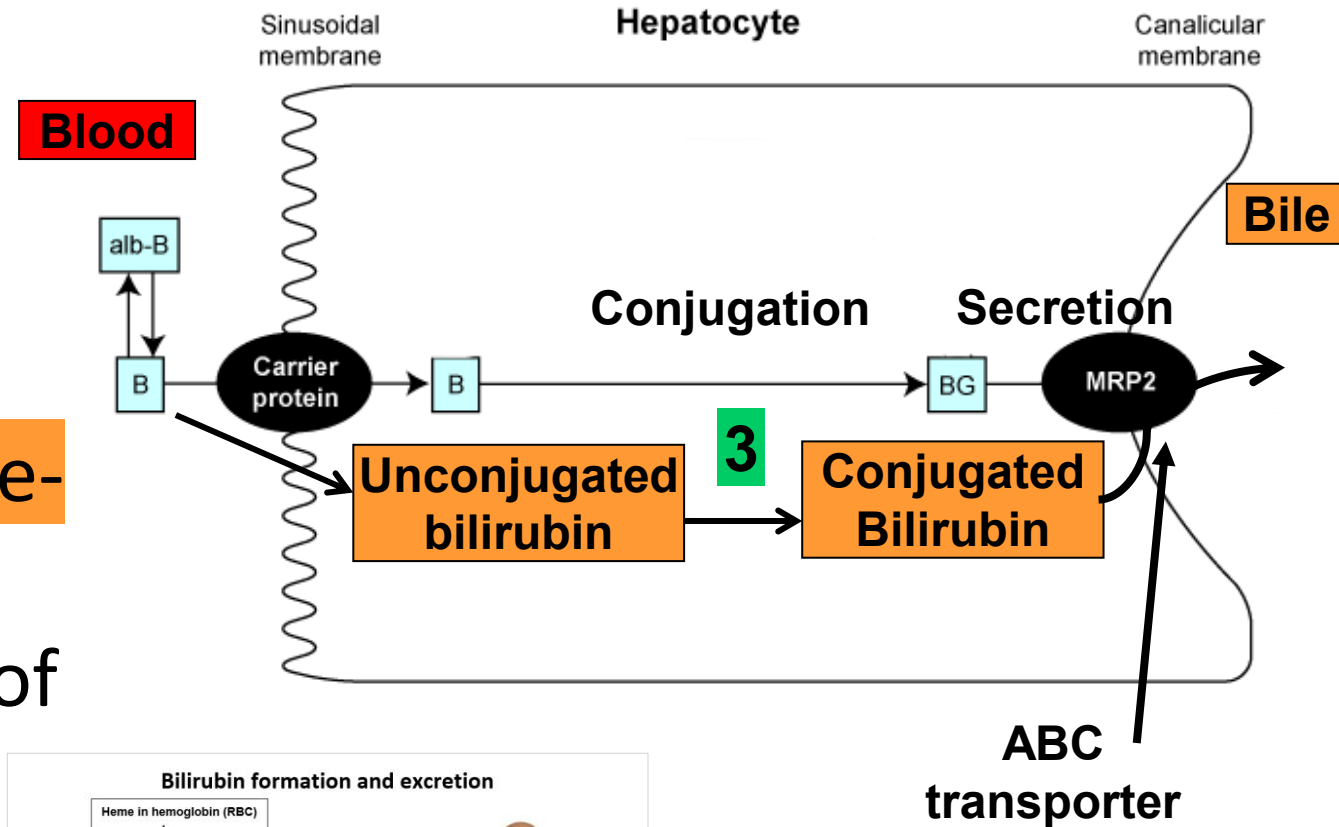
Conjugation of bilirubin in liver



- Unconjugated bilirubin converted to **conjugated bilirubin** (water soluble)
- Enzyme: **Bilirubin UDP-glucuronosyl transferase**
- Enzyme induction by **phenobarbital**

Secretion of conjugated bilirubin into bile

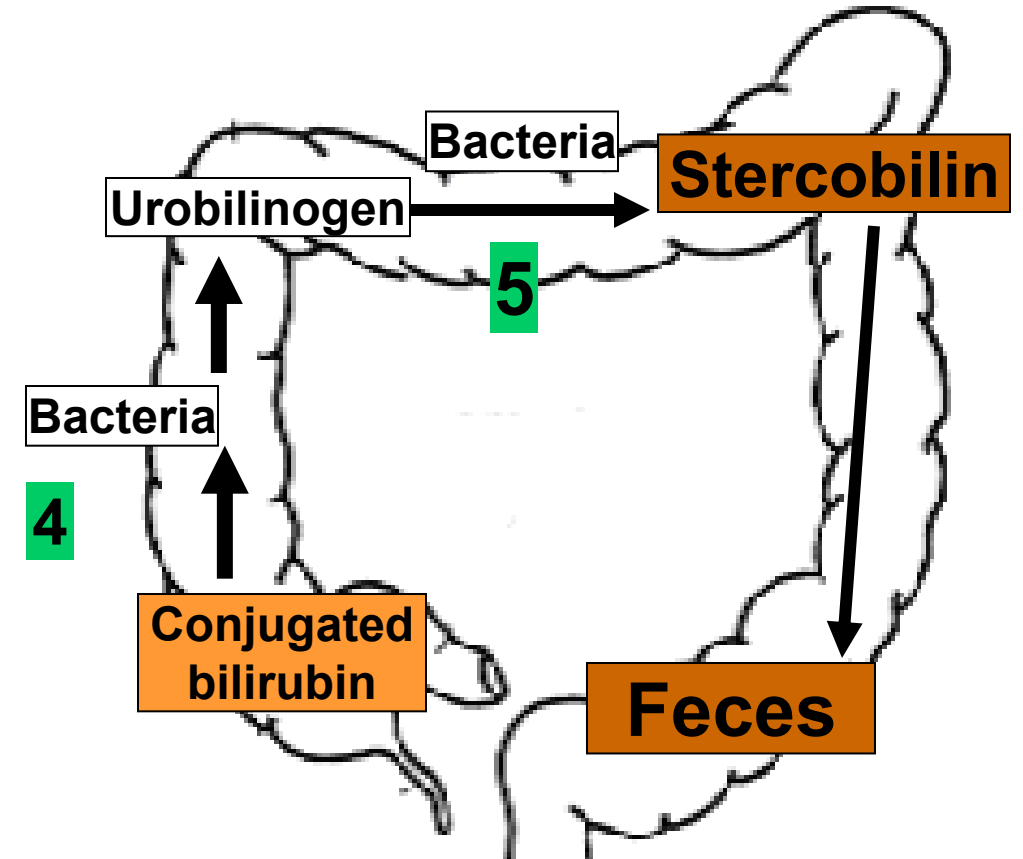
- Conjugated bilirubin secreted into bile canaliculus by specific **ABC transporters**
- Conjugated bilirubin (Bile pigment) contributes to **Orange-yellow color** of bile
- Bile released into second part of duodenum via common bile duct



**Defective in
Dubin-Johnson
syndrome**

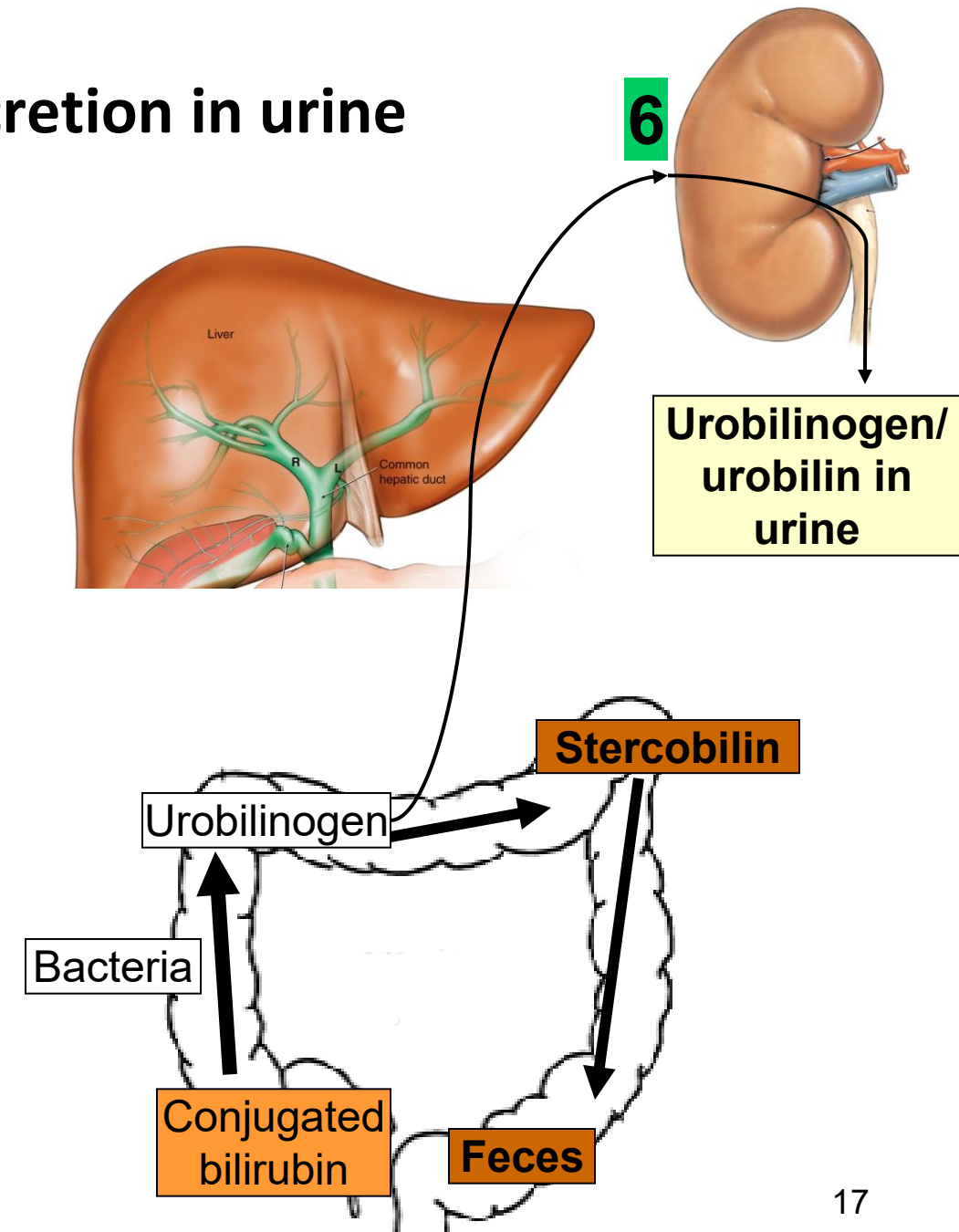
Formation of urobilinogen and stercobilin in large intestine

- **Bacterial flora:** Conjugated bilirubin forms **urobilinogen** (colorless)
- **Urobilinogen not same as unconjugated bilirubin**
- Urobilinogen forms **stercobilin**
- Stercobilin in feces **(brown color)**
- Some urobilinogen absorbed from gut into portal blood

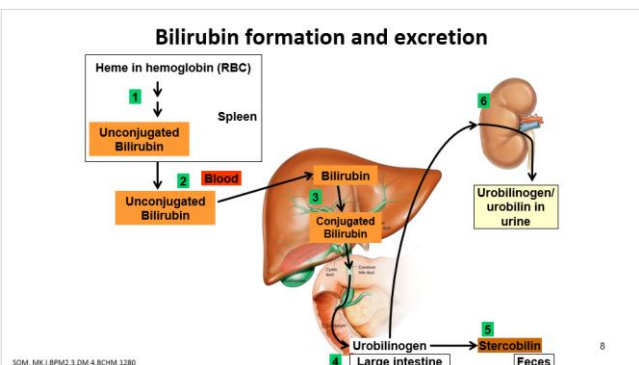
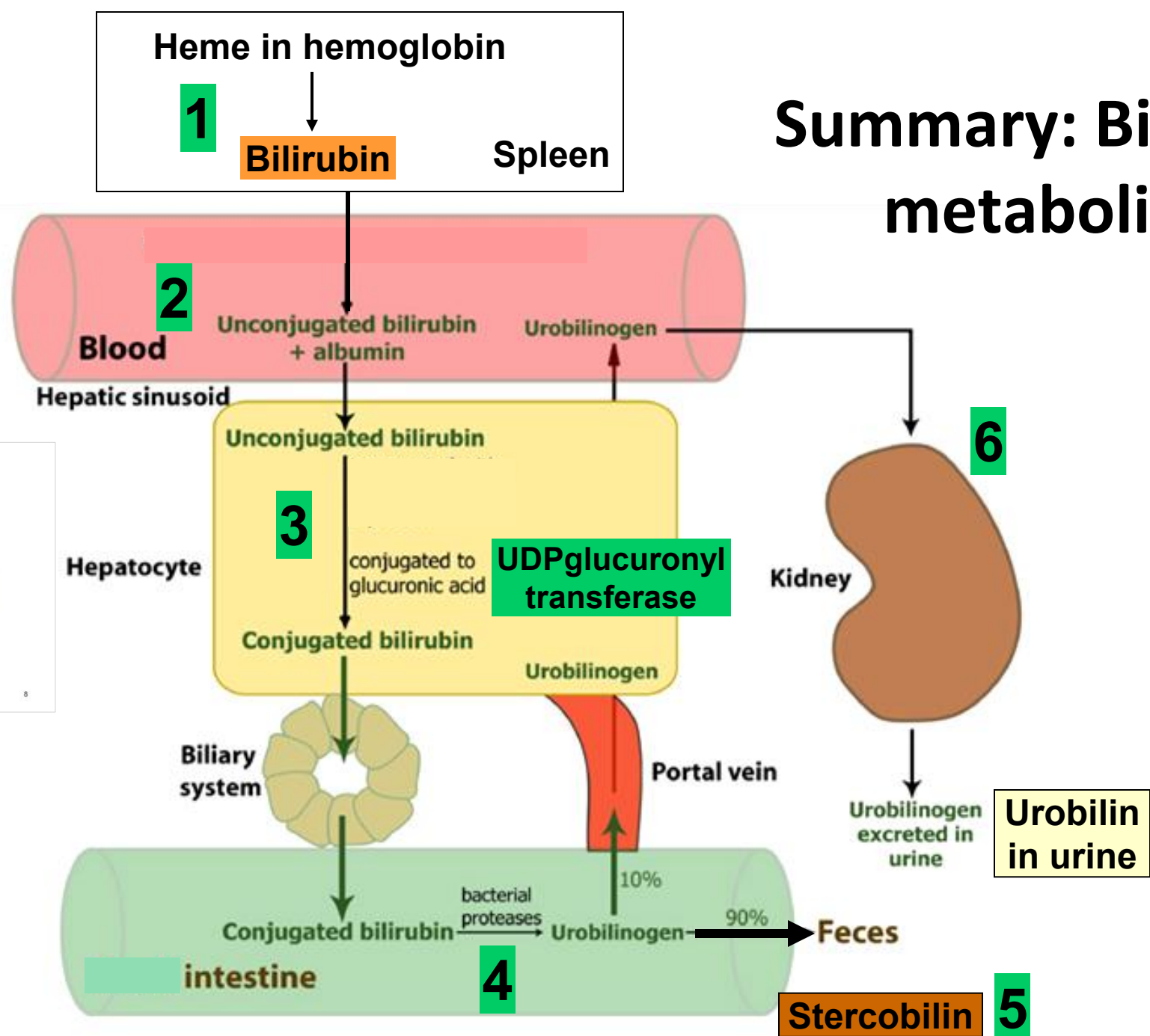


Urobilinogen: Absorption and excretion in urine

- 10% of urobilinogen in intestine absorbed into portal blood and goes to liver
- Small part re-excreted into bile canaliculus
- Majority of urobilinogen lost in urine urobilin (light yellow color)
- Urobilinogen and Urobilin synonymous

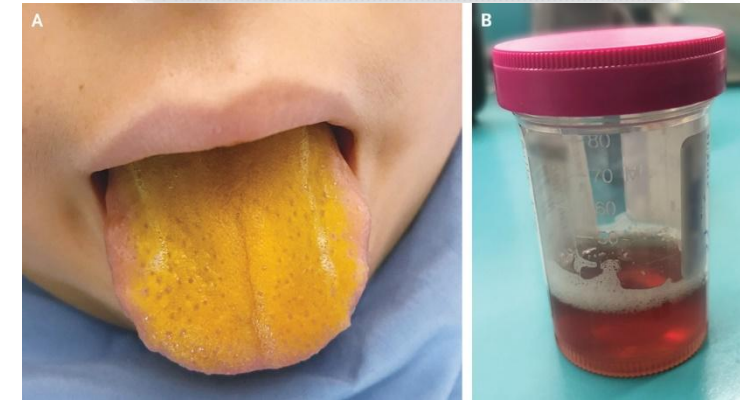


Summary: Bilirubin metabolism



Jaundice

- Yellowish discoloration of skin, mucous membranes, sclera, nail beds
- **Bilirubin (Bile pigment)** binds connective tissue
- Serum bilirubin levels more than 2 mg/dL (hyperbilirubinemia)
- Normal total serum bilirubin: (<1mg/dL)
 - Unconjugated (80%)
 - Conjugated (20%)
- **Normal URINE findings:**
 - Bilirubin absent;
 - Urobilinogen (urobilin) is present (Dipstick)

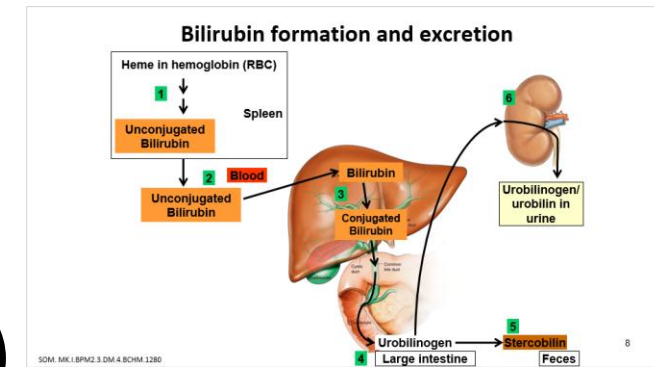


Lab tests to detect bilirubin fractions

- Van den Bergh reaction
 - Bilirubin with diazo reagent forms red colored complex
- Conjugated bilirubin (**direct reacting**) **water soluble** reacts rapidly with reagent
- Total bilirubin
 - Reaction in presence of methanol, both conjugated and unconjugated bilirubin react
- Unconjugated bilirubin (**indirect reacting**) **water insoluble** and reacts in presence of methanol
- Total bilirubin – **Direct (conjugated)** bilirubin = **Indirect (unconjugated)** bilirubin

Classification of jaundice (hyperbilirubinemia)

- Based on etiology (cause)
 - **Prehepatic (hemolytic)** jaundice: Increased breakdown of RBCs (Decreased lifespan of RBC)
 - **Hepatic jaundice**: Decreased conjugation capacity of liver and decreased excretion of bilirubin
 - **Posthepatic (obstructive/ cholestatic)** jaundice: Decreased excretion of bilirubin via bile duct

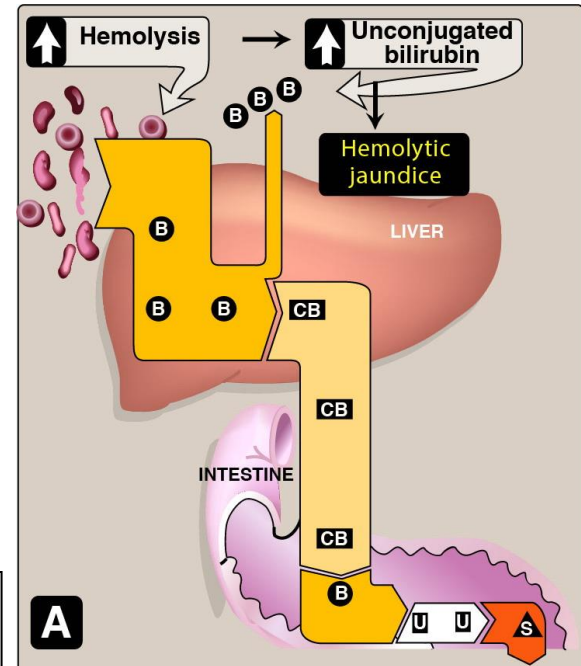
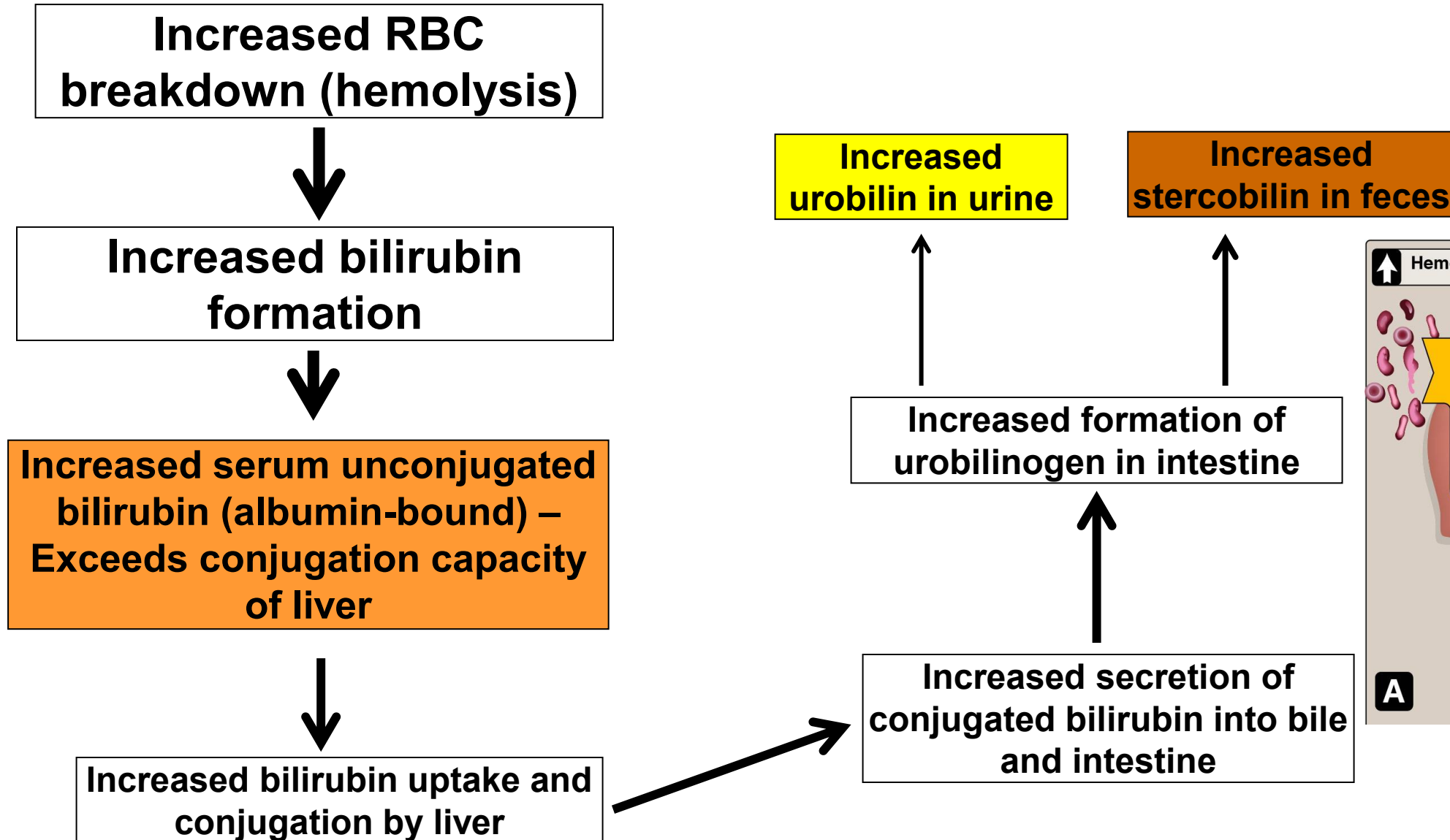


Prehepatic (Hemolytic) jaundice

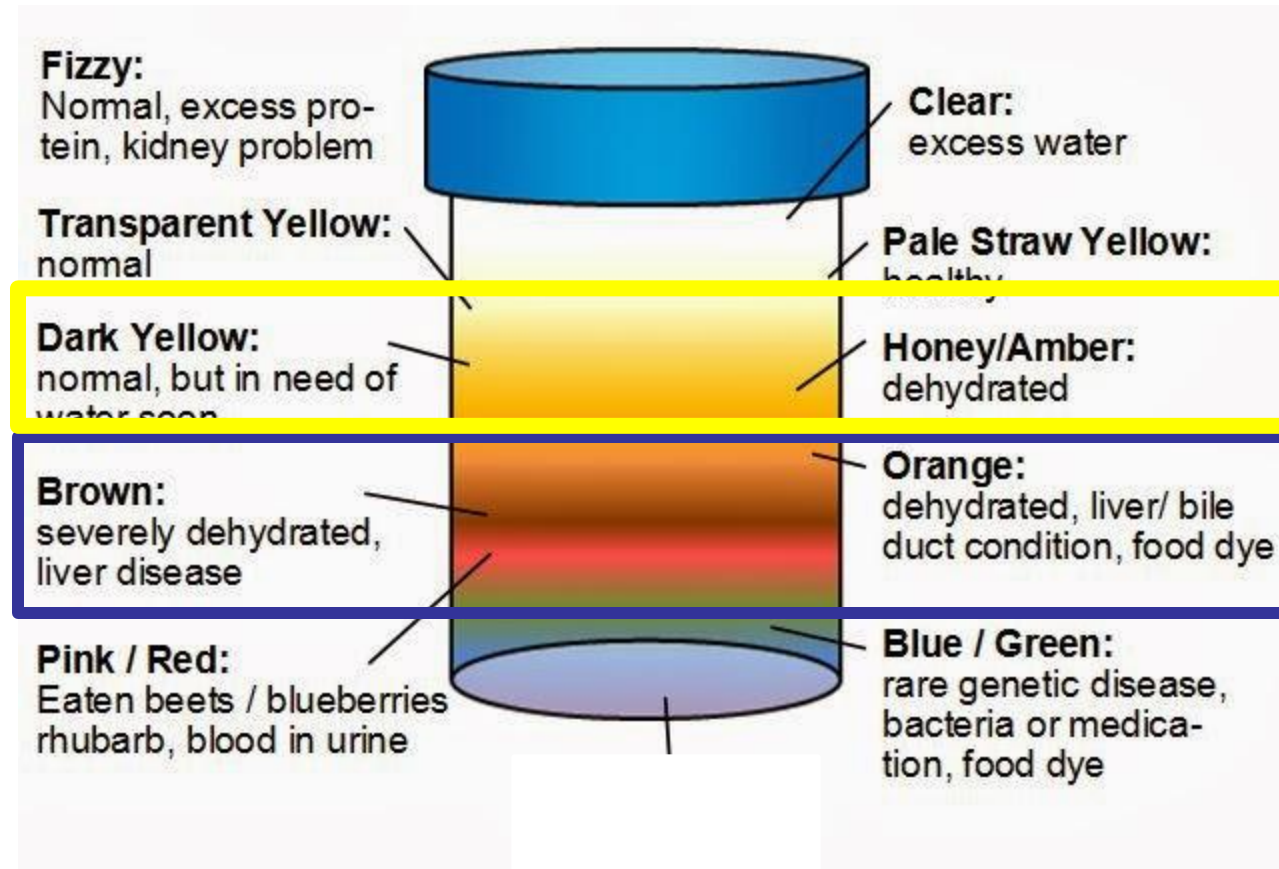
- Will Sichel has severe pain in back and abdomen (sickling crisis)
- On admission,
 - Hemoglobin = 7.8g/dL (normal: 12-16g/dL)
 - Serum total bilirubin = 3.5mg/dL
 - Serum conjugated (direct) bilirubin=0.2mg/dL
 - Serum lactate dehydrogenase (LDH): elevated
 - Serum haptoglobin: Low
 - Urine urobilinogen increased
 - Urine bilirubin absent
- What type of bilirubin is elevated in serum???



Changes in bilirubin metabolism in prehepatic (hemolytic) jaundice



Urine color prehepatic vs posthepatic jaundice



**Normal urine color/
Prehepatic jaundice**

**Hepatic and
Posthepatic jaundice**

Laboratory findings in prehepatic (hemolytic) jaundice

	Serum total bilirubin	Serum conjugated bilirubin	Serum unconjugated bilirubin	Urine bilirubin	Urine urobilinogen
Prehepatic jaundice	↑↑	N	↑↑	Absent (acholuric jaundice)	↑

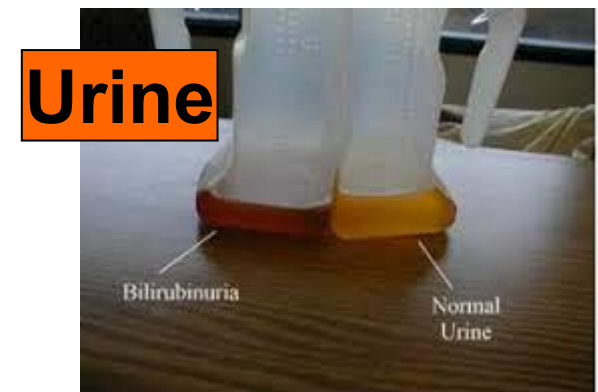
Why is bilirubin not found in urine in prehepatic jaundice???

Why is serum conjugated bilirubin not increased?

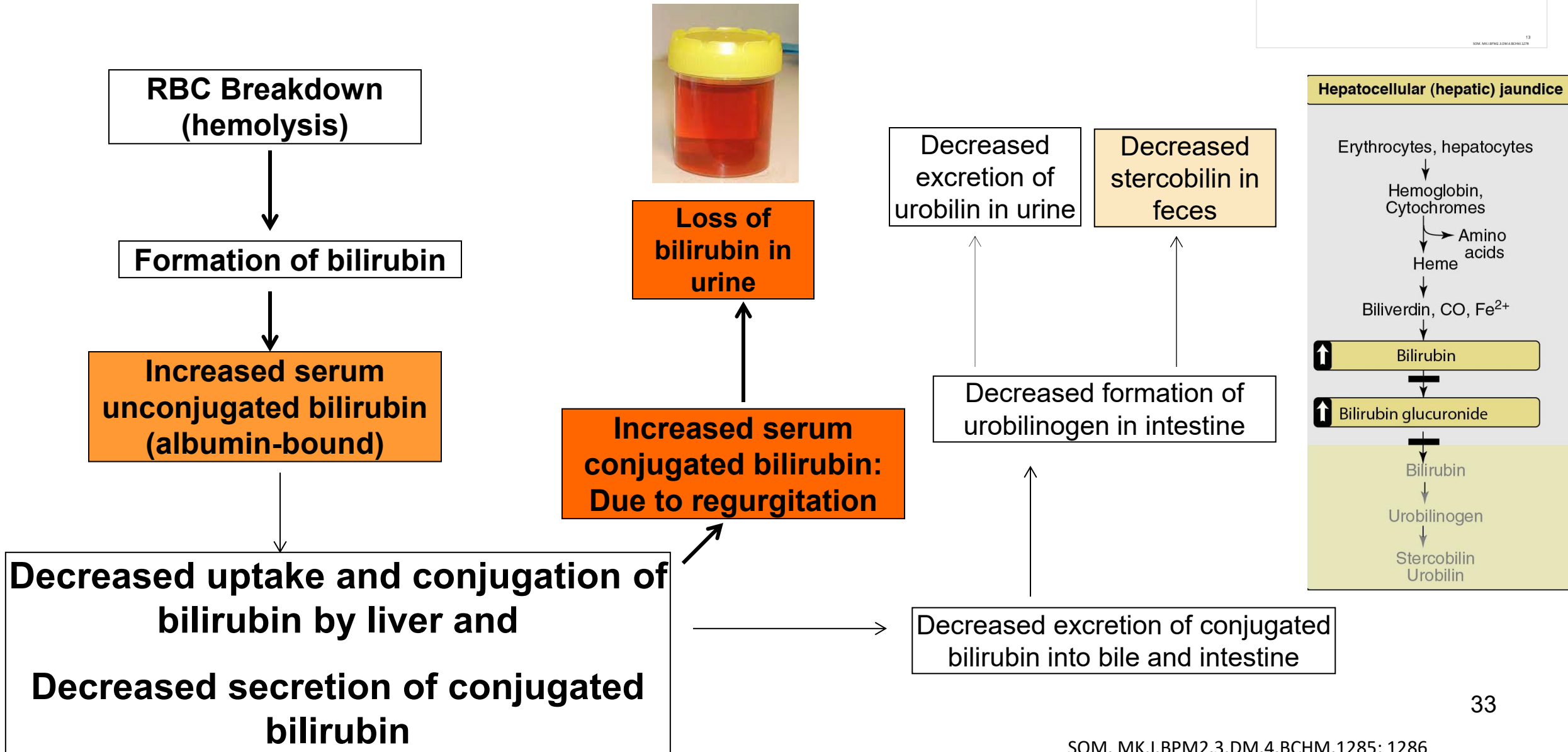
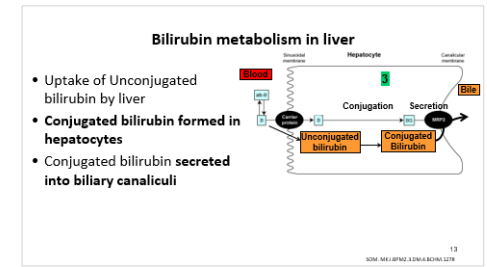
**Hemolytic anemia: G6PD deficiency; Alpha and beta thalassemias;
Any cause of hemolysis**

Hepatocellular jaundice

- A 40-year-old man has a 3-day history of fatigue and anorexia. Dark-colored urine
- On examination, scleral icterus (jaundice)
 - Serum total bilirubin=5mg/dL
 - Serum conjugated (direct) bilirubin=3mg/dL
 - Urine: orange yellow (positive for bilirubin)
- What type of bilirubin is elevated in serum?
- What type of bilirubin is found in urine??
- What other lab tests would you perform?



Changes in bilirubin metabolism in hepatic jaundice



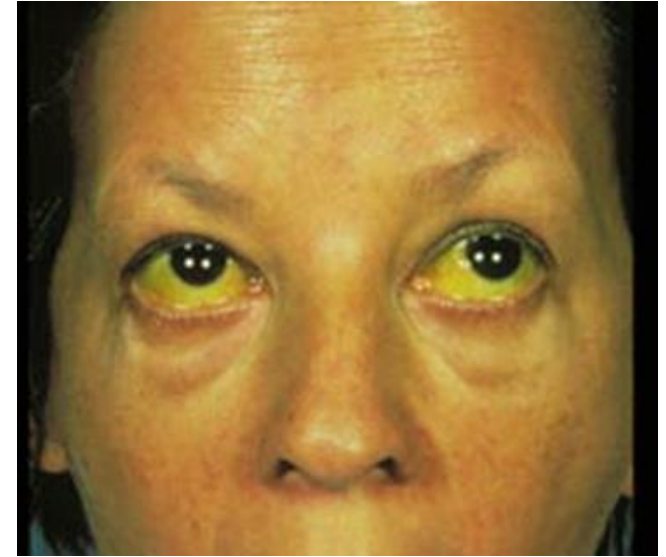
Laboratory findings in hepatic jaundice

	Serum total bilirubin	Serum conjugated bilirubin	Serum unconjugated bilirubin	Urine bilirubin	Urine urobilinogen
Hepatic jaundice	↑↑	↑	↑	Present	N or ↓ or ↑

- **Serum conjugated bilirubin** elevated: Reduced secretion of conjugated bilirubin into biliary canaliculus. Conjugated bilirubin regurgitates into plasma
- **Serum unconjugated bilirubin** elevated: Hepatocellular damage results in lower uptake of unconjugated bilirubin. Also, liver damage reduces conjugating capacity of liver.
- Additional lab findings: Elevated serum ALT and AST levels (refer Liver function tests lecture)

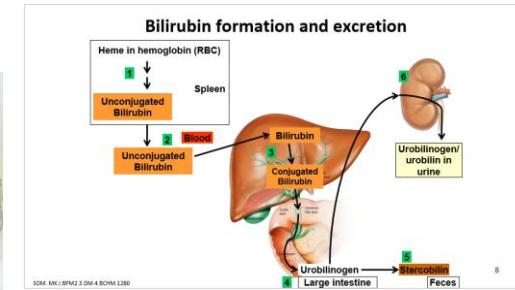
Posthepatic (obstructive/ cholestatic) jaundice

- A 50-year-old woman has a 2-month history of worsening jaundice.
- Severe itching: **Bile salts**
- Urine: Tea-colored for the last month.
- Good appetite; Lost 30 lbs in the past year. Why?
- Laboratory tests:
 - Serum total bilirubin 6mg/dL
 - Serum conjugated (direct) bilirubin 5.5mg/dL
 - Urine bilirubin positive
 - Urine urobilinogen absent
- What other lab tests would you perform?
- Why are stools **greasy** and pale colored?



Pale stools and dark-orange urine

Changes in bilirubin metabolism in posthepatic (obstructive/ cholestatic) jaundice



RBC Breakdown (hemolysis)



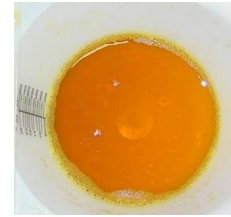
Formation of bilirubin



Serum Unconjugated bilirubin: Normal



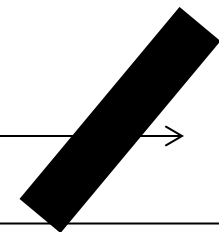
Normal uptake and conjugation of bilirubin by liver



Bilirubin in urine



Increased Serum conjugated bilirubin: Due to regurgitation

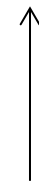


Bile duct obstruction due to stones in common bile duct or cancer

Urine urobilin: Absent



Decreased formation of urobilinogen in intestine

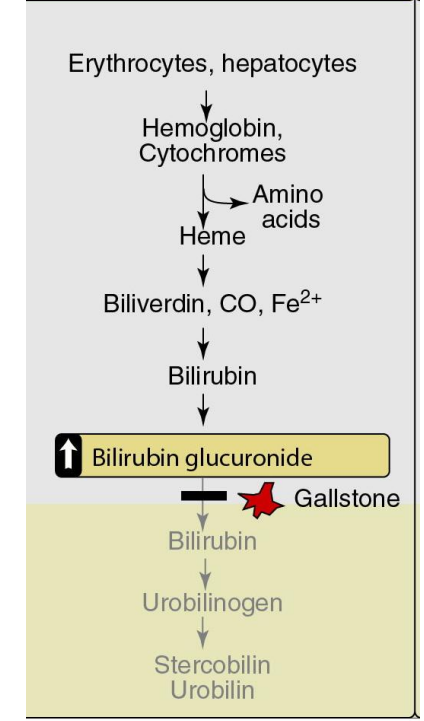


Fecal stercobilin: Decreased (clay-colored feces)



Decreased/absent conjugated bilirubin secretion into bile (Intestine)

Obstructive (posthepatic) jaundice



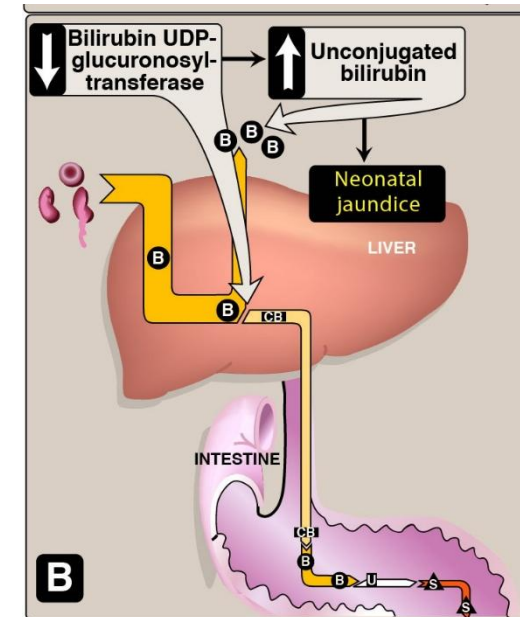
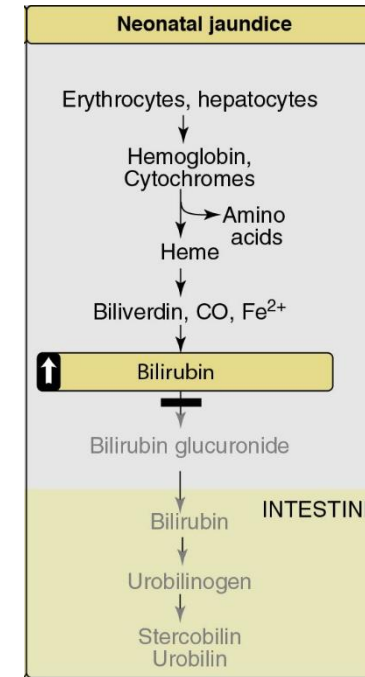
Laboratory findings in posthepatic (obstructive/ cholestatic) jaundice

	Serum total bilirubin	Serum conjugated bilirubin	Serum unconjugated bilirubin	Urine bilirubin	Urine urobilinogen
Posthepatic jaundice	↑↑	↑↑	N	Present	↓/ Absent

- Serum **conjugated bilirubin** elevated: Impaired secretion into biliary canaliculus due to obstruction (Cannot be excreted into bile)
 - **Conjugated bilirubin** regurgitates into plasma
- Urine urobilinogen: absent/ low (indicates extent of obstruction)
- Additional lab findings: Elevated serum ALP and GGT levels (refer Liver function tests lecture)

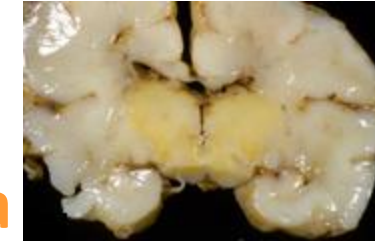
Physiological jaundice in newborn

- **Low** hepatic **UDP-glucuronosyl transferase** (premature infants more risk)
- Jaundice from day 2 and clears by 7th day
- Which fraction of bilirubin is increased??
Serum **Unconjugated bilirubin**
- Phototherapy (about 450nm)
- Light converts bilirubin to polar, water-soluble isomers (basis of bili-lights)



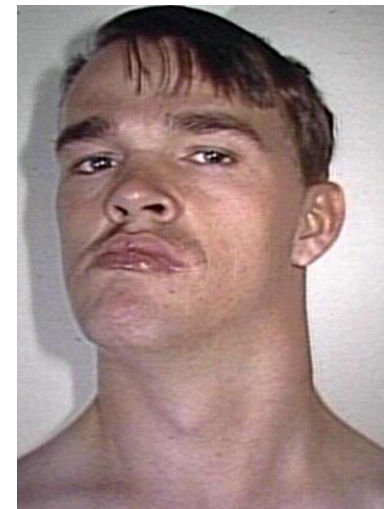
Jaundice in newborn: Kernicterus

- Serum bilirubin >25mg/dL
- **Unconjugated bilirubin** (Lipid soluble) **crosses blood-brain barrier** (not developed) deposits in **basal ganglia** of brain '**kernicterus**'
- Drugs (salicylates, sulfonamides...) displace unconjugated bilirubin (increased risk of kernicterus)
- Lethargy, altered muscle tone, high pitched cry
- Severe permanent neurologic symptoms (**choreoathetosis, spasticity, muscular rigidity, ataxia, intellectual impairment**)



Bilirubin moves from bloodstream into brain tissue

ADAM.



Case report on inherited hyperbilirubinemia

- A 2-day-old girl has severe **unconjugated hyperbilirubinemia**.
 - Phototherapy to reduce bilirubin levels
 - But unaffected by phenobarbital.
- **Crigler-Najjar syndrome type-I**: Traces (very low) of **bilirubin-UDP-glucuronyltransferase** activity in liver
- Stress/illness precipitates kernicterus (slurred speech, ataxia, and coma)
 - Treat with antibiotics, plasmapheresis, and **intensive phototherapy**;
 - Recovery without neurologic sequelae
- 10-12 hours daily phototherapy; Serum bilirubin levels at **24 to 27 mg/dL**
- Liver transplantation

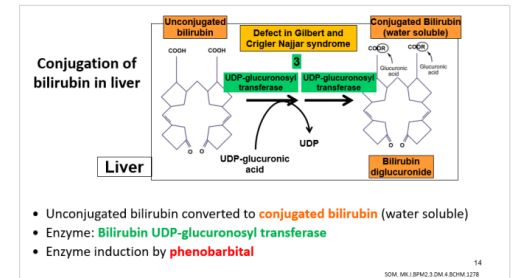
Modified from NEJM, 1998

<https://www.youtube.com/watch?v=bBgfPsMotnw>

Inherited **Unconjugated** hyperbilirubinemia

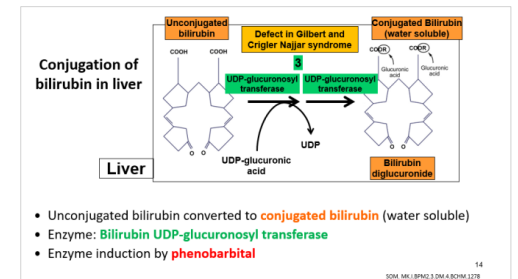
- Crigler-Najjar syndromes I and II, Gilbert syndrome
- Varying deficiency of **bilirubin UDP-glucuronyl transferase**
- **Unconjugated hyperbilirubinemia**

- **Crigler-Najjar syndrome I**
- Most severe; Very low (traces) enzyme activity (<5%)
- Jaundice in infancy
- Serum bilirubin levels up to 50 mg/dL (**Unconjugated**)
- Kernicterus and developmental delay
- Daily phototherapy to prevent kernicterus
- Do NOT respond to phenobarbital
- Usually fatal (if not treated) due to kernicterus



Inherited **Unconjugated** hyperbilirubinemia

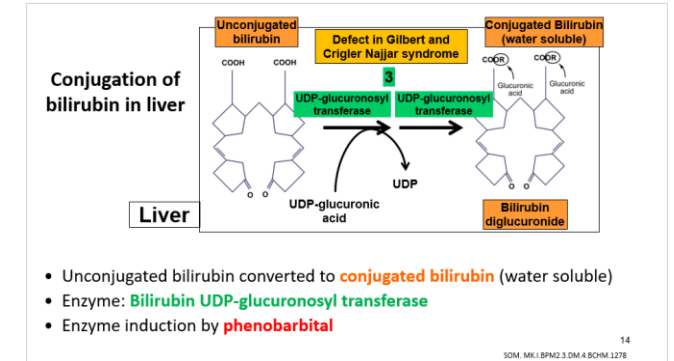
- **Crigler-Najjar syndrome II (Arias syndrome)**
 - Lower activity of **bilirubin glucuronyl transferase** (10-20%)
 - Jaundice, but less severe than type I (serum **unconjugated bilirubin** levels 6-22mg/dL)
 - Respond to **phenobarbital** (induces enzyme)
 - Regular phototherapy



Inherited **Unconjugated** hyperbilirubinemia

- **Gilbert syndrome**

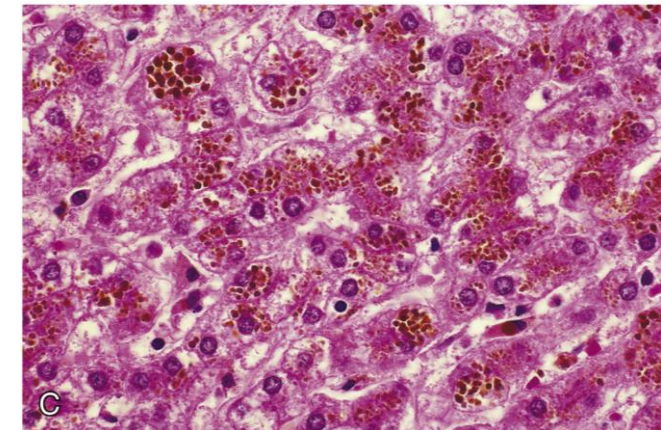
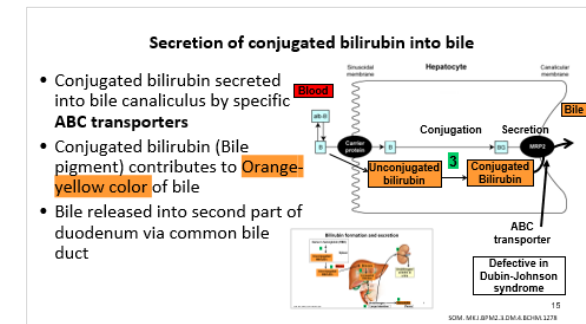
- Present in 3-7% of population (common)
- Adolescents and young adults
- Mild jaundice (2-5 mg/dL) following **illness or stress or starvation**
- **Bilirubin UDP-glucuronyl transferase: 50% activity**
- Mild increase in serum **unconjugated (indirect) bilirubin**
- **Normal AST, ALT and ALP levels**



Inherited **Conjugated** hyperbilirubinemia

- **Dubin-Johnson syndrome**

- Inherited deficiency of **ABC transporter** (secretes conjugated bilirubin into biliary canaliculus)
- Adolescents and young adults
- Elevated serum **conjugated (direct) bilirubin**
- **Normal AST, ALT and ALP levels**
- Liver appears **black** (gross appearance)
- Liver biopsy: **Pigments** in hepatocytes



Summary table of inherited hyperbilirubinemia

Disorder	Age of onset	Degree of enzyme deficiency	Type of bilirubin in serum
Crigler Najjar syndrome type I			
Crigler Najjar syndrome type II			
Gilbert syndrome			
Dubin Johnson syndrome			

Summary of serum and urine findings in Jaundice

	Serum total bilirubin	Serum conjugated (Direct) bilirubin	Serum unconjugated (Indirect) bilirubin	Urine bilirubin	Urine urobilinogen
Prehepatic/ Hemolytic jaundice	↑↑	N	↑↑	Absent (acholuric jaundice)	↑
Hepatic/ Hepatocellular jaundice	↑↑	↑	↑	Present	N or ↓ or ↑
Posthepatic/ Obstructive/ Cholestatic jaundice	↑↑	↑↑	N	Present	↓ / Absent

Summary: Heme synthesis;
Heme degradation and jaundice
Concept map: Fig 21.20

Remember, bile salts/bile acids (required for lipid digestion and absorption) are different from bile pigments/ bilirubin (excretory waste products of bilirubin metabolism) – both are components of BILE

